



VU NET

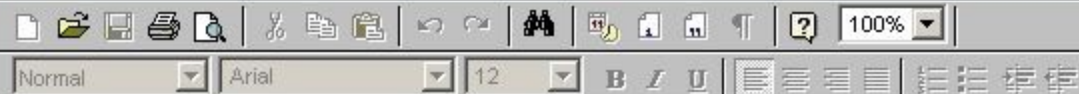


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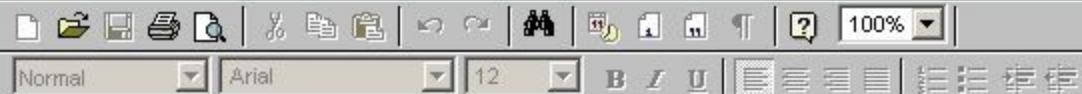
In which sciences or subjects Logistic equations are useful?

Answer ([Please click here to Add Answer](#))



Define cycle of a vibrating body?

Answer ([Please click here to Add Answer](#))



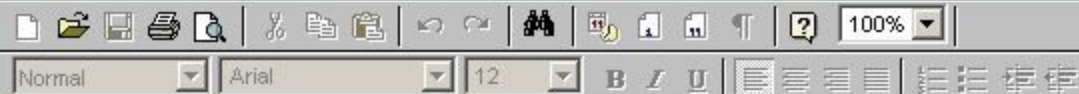
What is Logistic equation, write it mathematically?

Answer ([Please click here to Add Answer](#))



Write the solution $x(t) = c_1 \cos \omega t + c_2 \sin \omega t$ of a simple harmonic motion in a simple alternative form?

Answer ([Please click here to Add Answer](#))



Find the general solution of the following non-homogenous differential equation using annihilator operator

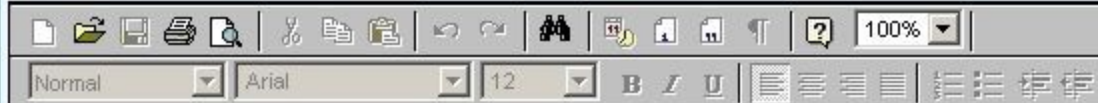
$$y'' - 3y' + 2y = 4x^2 + 5?$$

Answer ([Please click here to Add Answer](#))



Given that $y = c_1 e^x + c_2 e^{-x}$ is a two parameter family of solutions of the differential equation $\frac{d^2 y}{dx^2} - y = 0$ on $(-\infty, \infty)$ find a member of the family satisfying the boundary conditions $y(0) = 0, y'(1) = 1$

Answer ([Please click here to Add Answer](#))





In which sciences or subjects Logistic equations are useful?

Answer ([Please click here to Add Answer](#))





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What is Logistic equation, write it mathematically?

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Answer ([Please click here to Add Answer](#))



Made By: Waqar Siddhu

Given that $y = c_1 e^x + c_2 e^{-x}$ is a two parameter family of solutions of the differential equation $\frac{d^2 y}{dx^2} - y = 0$ on $(-\infty, \infty)$ find a member of the family satisfying the boundary conditions $y(0) = 0, y'(1) = 1$

Answer ([Please click here to Add Answer](#))



Question No : 21 of 26

Marks: 2 (Budgeted Time 4 Min) 

Find the order of the homogenous equation obtained from the following non-homogenous differential equation

$$y'' + 4y' + 3y = 4x^2 + 5?$$

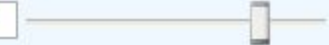
Answer ([Please click here to Add Answer](#)) 

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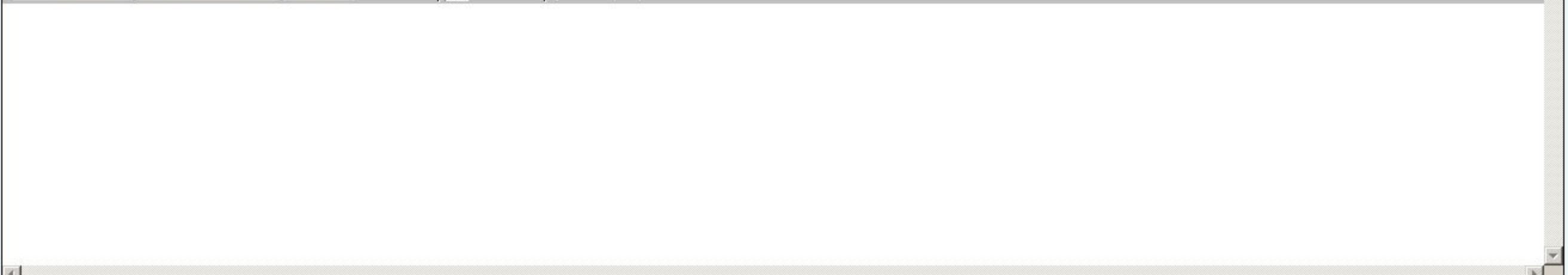
Question No : 22 of 26

Marks: 2 (Budgeted Time 4 Min) 

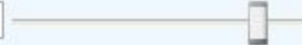
If half-time life (T) of a radioactive isotope is 5 days, then find k.

Answer ([Please click here to Add Answer](#)) 

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Question No : 23 of 26

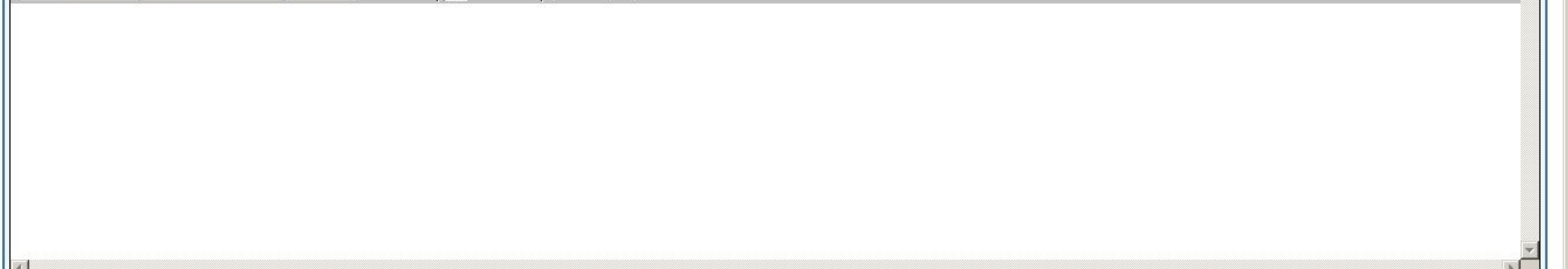
Marks: 3 (Budgeted Time 6 Min) 

Determine the annihilator operator of the general solution of the following homogeneous linear first order differential equation:

$$\frac{dy}{dx} - 5y = 0$$

Answer ([Please click here to Add Answer](#)) 

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Question No : 24 of 26

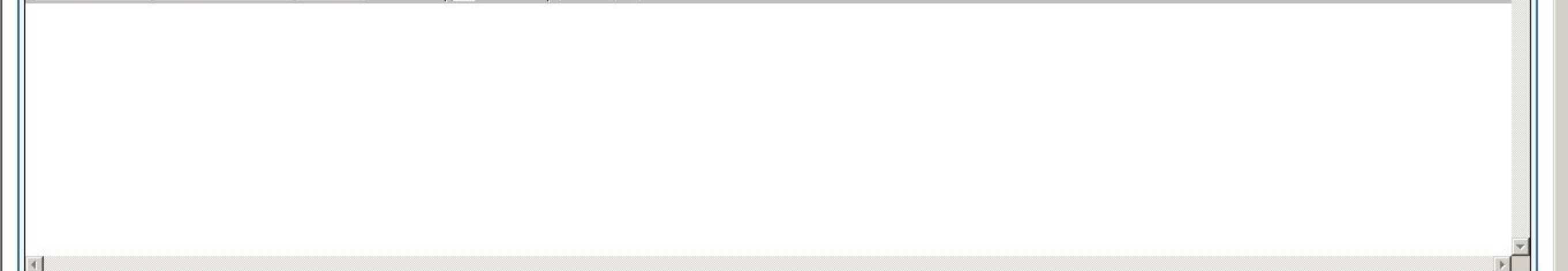
Marks: 3 (Budgeted Time 6 Min)

If $y_1(x) = e^x$ is the first solution of $\frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} - 3y = 0$, then construct its second solution using $y_2(x) = y_1(x) \int \frac{e^{-\int P(x) dx}}{y_1^2(x)} dx$.

Note: where $P(x)$ is a coefficient of the first derivative involving in the above given differential equation.

Answer ([Please click here to Add Answer](#))

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Question No : 25 of 26

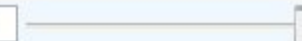
Marks: 5 (Budgeted Time 10 Min) 

Find period of vibration of the solution of a simple harmonic motion?

Answer ([Please click here to Add Answer](#)) 

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Question No : 26 of 26

Marks: 5 (Budgeted Time 10 Min)

Using Newton's Law of Cooling with initial condition $T(0)=T_0$ and if temperatures at time t_1 and t_2 are known then show that

$$\frac{T(t_1) - T_m}{T(t_2) - T_m} = e^{k(t_1 - t_2)}$$

Answer (Please click here to Add Answer)

If initial temperature of the cooling body is T_0 then we obtain the initial value problem

$$\frac{dT}{dt} = k(T - T_m), \quad T(0) = T_0$$

where k is constant of proportionality. The differential equation in the problem is linear as well as separable.

Separating the variables and integrating we obtain

$$\int \frac{dT}{T - T_m} = \int k dt$$

This means that

$$\ln |T - T_m| = kt + C$$

$$T - T_m = e^{kt+C}$$

$$T(t) = T_m + C_1 e^{kt} \quad \text{where } C_1 = e^C$$

Now applying the initial condition $T(0) = T_0$, we see that $C_1 = T_0 - T_m$. Thus the solution of the initial value problem is given by

$$T(t) = T_m + (T_0 - T_m)e^{kt}$$

Hence, If temperatures at times t_1 and t_2 are known then we have

$$T(t_1) - T_m = (T_0 - T_m)e^{kt_1}, \quad T(t_2) - T_m = (T_0 - T_m)e^{kt_2}$$

So that we can write

$$\frac{T(t_1) - T_m}{T(t_2) - T_m} = e^{k(t_1 - t_2)}$$

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Evaluate $e^{-\int p(x) dx}$ using the differential equation $x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + \left(x^2 - \frac{1}{4}\right)y = 0$ by writing it in the standard form $\frac{d^2 y}{dx^2} + P(x) \frac{dy}{dx} + Q(x)y = 0$.

Answer ([Please click here to Add Answer](#))



If the complementary solution of the following differential equation is $c_1 \sin 2x + c_2 \cos 2x$, then what will be the general form of its particular solution:

$$\frac{d^2 y}{dx^2} + 4y = \cos 2x$$

Answer ([Please click here to Add Answer](#))



If a mass weighing $8lb$ and $k = 40 lb/ft$ then find the amount of elongation's'.

Answer ([Please click here to Add Answer](#))



If $y_1(x) = e^{-4x}$ is the first solution of $\frac{d^2 y}{dx^2} + 6 \frac{dy}{dx} + 8y = 0$, then construct its second solution using $y_2(x) = y_1(x) \int \frac{e^{-\int P(x) dx}}{y_1^2(x)} dx$.

Note: where $P(x)$ is a coefficient of the first derivative involving in the above given differential equation.

Answer ([Please click here to Add Answer](#))



A mass weighing $2lb$ stretches a spring 6 inches. Determine the equation of simple harmonic motion.

Answer ([Please click here to Add Answer](#))



A radioactive isotope has a half-life of 16 days. We have 45 g at the end of 32 days. How much radioisotope was initially present?

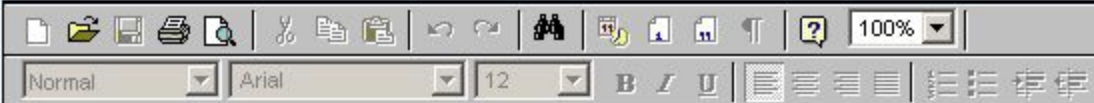
Answer ([Please click here to Add Answer](#))



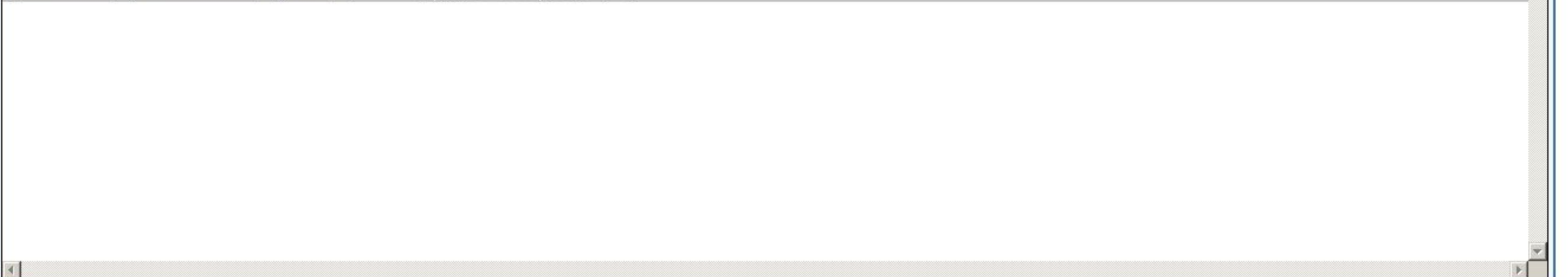
Question No : 21 of 26

Marks: 2 (Budgeted Time 4 Min) 

If a mass weighing $20lb$ stretches a spring by $\frac{1}{2}ft$ then find k using the Hooke's Law.

Answer ([Please click here to Add Answer](#)) 

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21 

If a mass weighing $20lb$ stretches a spring by $\frac{1}{2}ft$ then find k using the Hooke's Law.

Answer ([Please click here to Add Answer](#))



Question No : 22 of 26

Marks: 2 (Budgeted Time 4 Min)

Show that $y = x^2$ is a solution of the following differential equation:

$$x^2 \frac{d^2 y}{dx^2} - 3x \frac{dy}{dx} + 4y = 0$$

Answer ([Please click here to Add Answer](#))

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22



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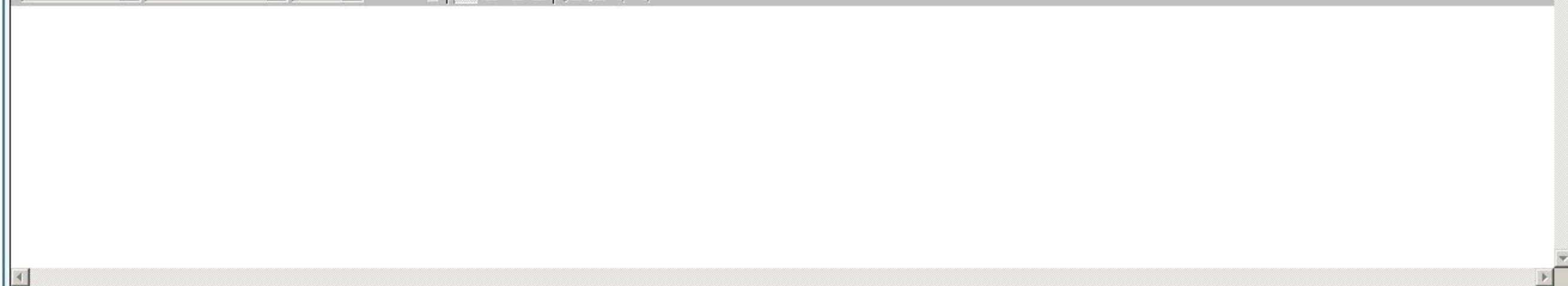
Question No : 23 of 26

Marks: 3 (Budgeted Time 6 Min) 

What is Logistic equation, write it mathematically?

Answer ([Please click here to Add Answer](#)) 

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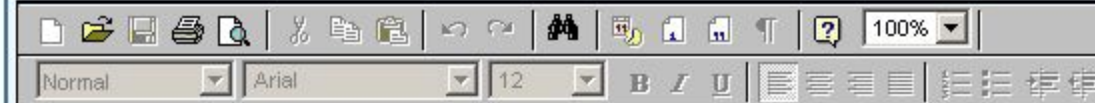


Question No : 24 of 26

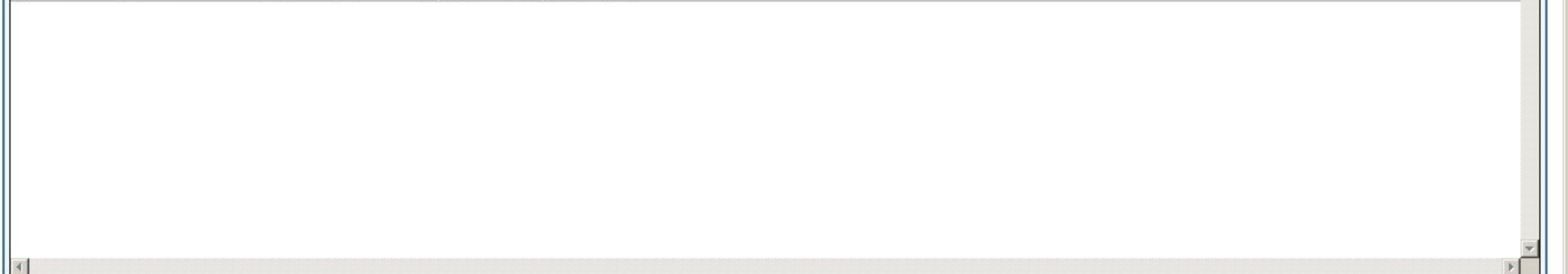
Marks: 3 (Budgeted Time 6 Min) 

Write the following differential equation in the form $L(y) = g(x)$, where L is a linear differential operator with constant coefficients.

$$\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 5y = 2\sin y$$

Answer ([Please click here to Add Answer](#)) 

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Start Time: 6:56 AM

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Write the following differential equation in the form $L(y) = g(x)$, where L is a linear differential operator with constant coefficients.

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Answer ([Please click here to Add Answer](#))



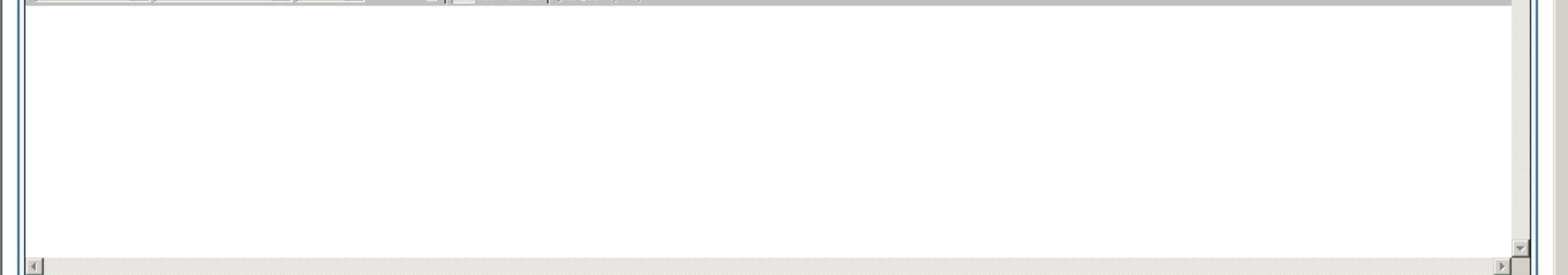
Question No : 25 of 26

Marks: 5 (Budgeted Time 10 Min) 

Find the wronskian of the Differential equation $y''' - 2y'' - 21y' - 18y = 3 + 4e^{-t}$ using variation of parameter and the root of the auxiliary equation is $m_1 = -3, m_2 = -1, m_3 = 6$?

Answer ([Please click here to Add Answer](#)) 

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25 

Find the wronskian of the Differential equation $y''' - 2y'' - 21y' - 18y = 3 + 4e^{-x}$ using variation of parameter and the root of the auxiliary equation is $m_1 = -3, m_2 = -1, m_3 = 6$?

Answer ([Please click here to Add Answer](#))



Question No : 26 of 26

Marks: 5 (Budgeted Time 10 Min) 

Solve the boundary value problem $\frac{d^2y}{dx^2} + 25y = 0$ subject to the boundary conditions $y(0) = 0$, $y\left(\frac{\pi}{10}\right) = 0$

Answer ([Please click here to Add Answer](#)) 

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26 

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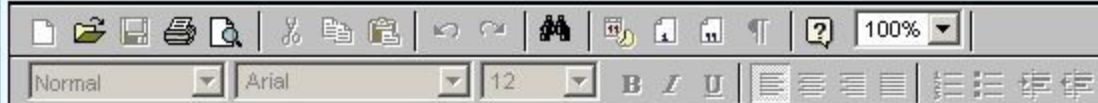
Answer ([Please click here to Add Answer](#))



Find the order of the homogenous equation obtained from the following non-homogenous differential equation

$$y'' + 4y' + 3y = 4x^2 + 5?$$

Answer ([Please click here to Add Answer](#))



If half-time life (T) of a radioactive isotope is 5 days, then find k.

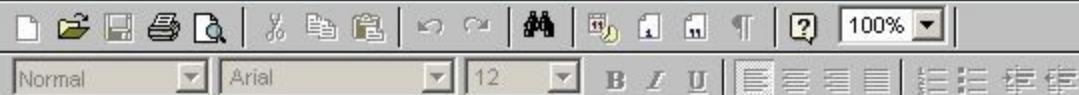
Answer ([Please click here to Add Answer](#))



Determine the annihilator operator of the general solution of the following homogeneous linear first order differential equation:

$$\frac{dy}{dx} - 5y = 0$$

Answer ([Please click here to Add Answer](#))



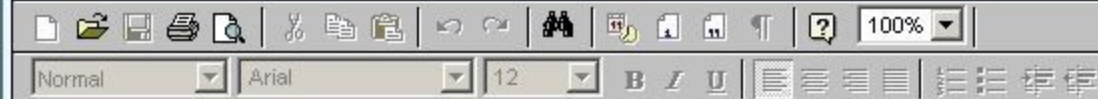
If $y_1(x) = e^x$ is the first solution of $\frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} - 3y = 0$, then construct its second solution using $y_2(x) = y_1(x) \int \frac{e^{-\int P(x) dx}}{y_1^2(x)} dx$.

Note: where $P(x)$ is a coefficient of the first derivative involving in the above given differential equation.

Answer ([Please click here to Add Answer](#))



Answer (Please [click here](#) to Add Answer)



Using Newton's Law of Cooling with initial condition $T(0)=T_0$ and if temperatures at time t_1 and t_2 are known then show that

$$\frac{T(t_1) - T_m}{T(t_2) - T_m} = e^{k(t_1 - t_2)}$$

Answer ([Please click here to Add Answer](#))



Show that the functions $f_1(x) = \sin x$ and $f_2(x) = x$ are linearly independent.

Answer ([Please click here to Add Answer](#))



If the complementary solution of the following differential equation is $c_1 \sin 2x + c_2 \cos 2x$, then what will be the general form of its particular solution:

$$\frac{d^2 y}{dx^2} + 4y = \cos 2x$$

Answer ([Please click here to Add Answer](#))



Calculate the Wronskian of the following functions. Further determine whether they are linearly independent or not on the interval $(-\infty, \infty)$.

$$f_1(x) = \sin^2 x ; f_2(x) = \cos^2 x$$

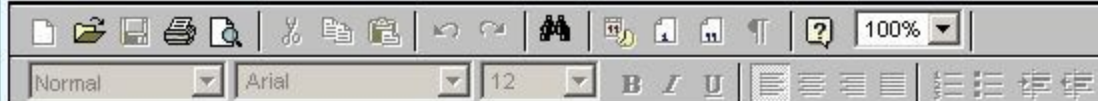
Answer ([Please click here to Add Answer](#))



What will be the general form of the particular solution of the following differential equation:

$$\frac{d^2y}{dx^2} - 2 \frac{dy}{dx} + y = e^x$$

Answer ([Please click here to Add Answer](#))



Deduce the Special Case of Logistic Equation "Epidemic spread"?

Answer ([Please click here to Add Answer](#))



Find the general solution of the following non-homogenous differential equation using annihilator operator

$$y'' - 3y' + 2y = 4x^2 + 5$$

Answer ([Please click here to Add Answer](#))



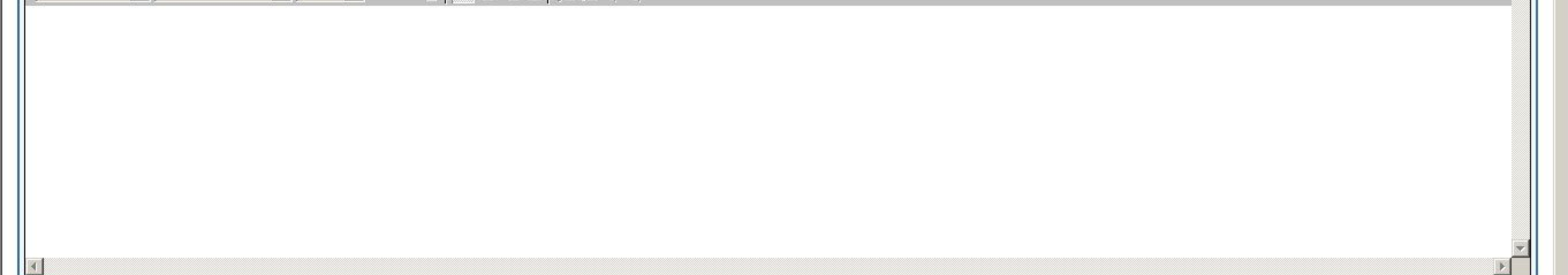
Question No : 21 of 26

Marks: 2 (Budgeted Time 4 Min) 

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Answer ([Please click here to Add Answer](#)) 

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Question No : 22 of 26

Marks: 2 (Budgeted Time 4 Min)

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Answer ([Please click here to Add Answer](#))

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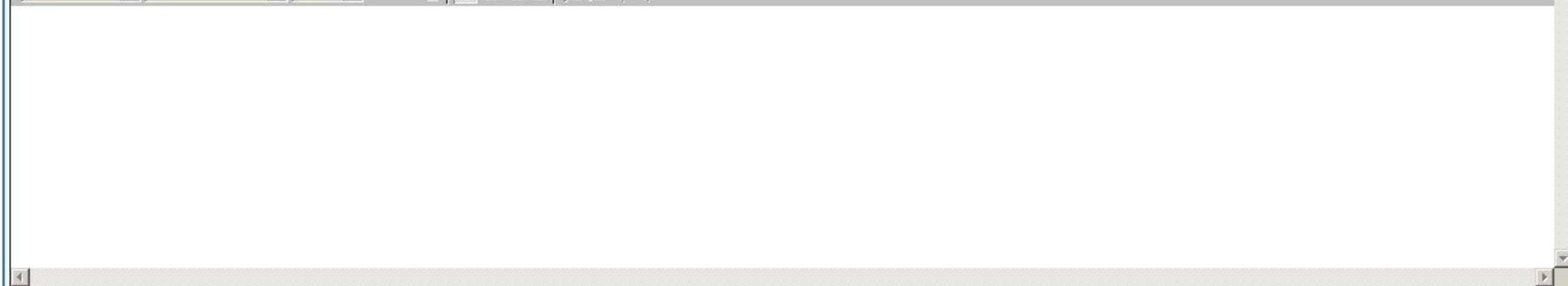
Question No : 23 of 26

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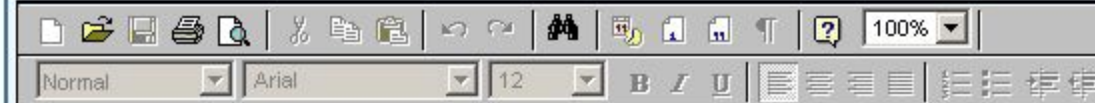
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Question No : 24 of 26

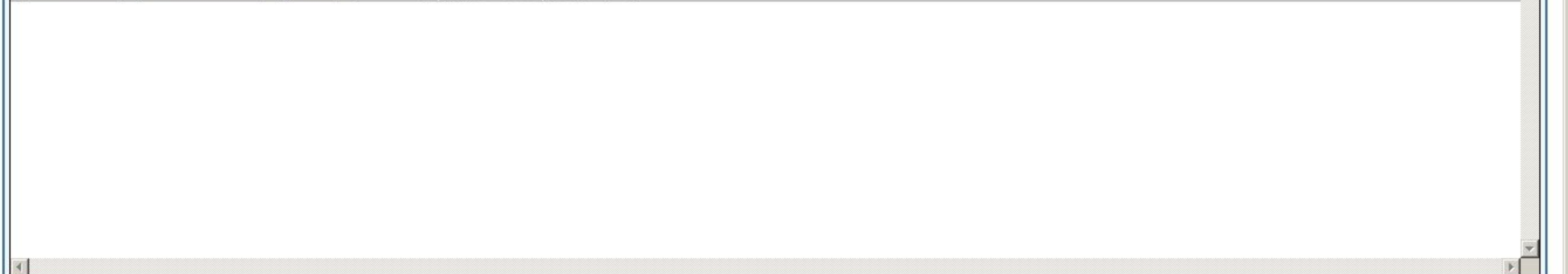
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$$\frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} + 5y = 2 \sin y$$

Answer ([Please click here to Add Answer](#)) 

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Start Time: 6:56 AM

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Time Left 24 

Question No : 25 of 26

Marks: 5 (Budgeted Time 10 Min) 

Find the wronskian of the Differential equation $y''' - 2y'' - 21y' - 18y = 3 + 4e^{-t}$ using variation of parameter and the root of the auxiliary equation is $m_1 = -3, m_2 = -1, m_3 = 6$?

Answer ([Please click here to Add Answer](#)) 

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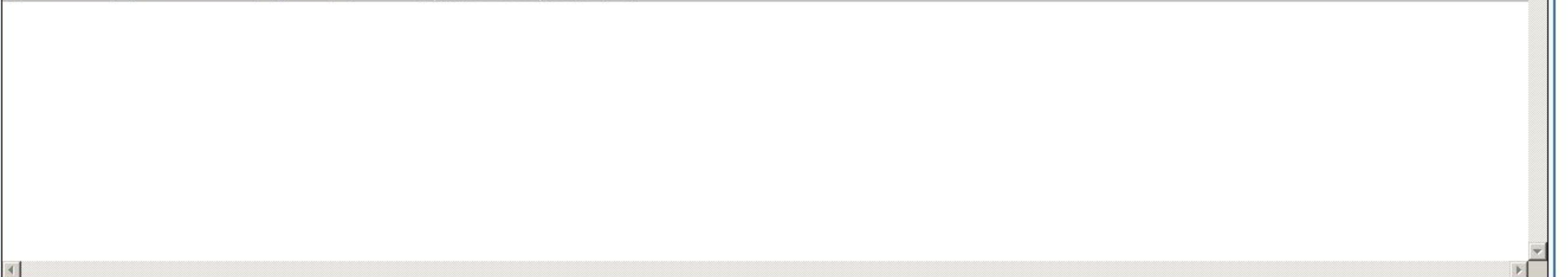
Question No : 26 of 26

Marks: 5 (Budgeted Time 10 Min) 

Solve the boundary value problem $\frac{d^2y}{dx^2} + 25y = 0$ subject to the boundary conditions $y(0) = 0$, $y\left(\frac{\pi}{10}\right) = 0$

Answer ([Please click here to Add Answer](#)) 

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Start Time: 6:56 AM

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Time Left 26 

Differential Equations (MTH401)

Question: **21** (Marks: 2)Attempted Questions: **25** Total Questions: **26**

Determine whether the following function is homogeneous or not? If not then justify the answer with appropriate reason:

$$f(x, y) = \frac{1}{y}$$

Differential Equations (MTH401)

Question: **22** (Marks: 2)Attempted Questions: **25** Total Questions: **26**

Find the characteristic (auxiliary) equation and its roots for the following homogeneous linear differential equation,

$$\frac{d^2 y}{dx^2} - 9y = 0$$

Answer:

Differential Equations (MTH401)

Question: **23** (Marks: 3)Attempted Questions: **25** Total Questions: **26**

A mass weighing 2 lb stretches a spring 6 inches. Determine the equation of the simple harmonic motion.

.....

Differential Equations (MTH401)

Question: **24** (Marks: 3)Attempted Questions: **25** Total Questions: **26**

Find the differential equation of the orthogonal family of curves $y = cx^2$, where c is an arbitrary constant.

.....

Answer:

Differential Equations (MTH401)

Question: **25** (Marks: 5)Attempted Questions: **25** Total Questions: **26**

Discuss the exactness of the following differential equation and solve it
 $(\sin y - y \sin x) dx + (\cos x + x \cos y) dy = 0$

.....

Answer:

Differential Equations (MTH401)

Question: **26** (Marks: 5)Attempted Questions: **25** Total Questions: **26**

Given $L = D^2 + 5D + 6$ is a linear differential operator. Then

1. Factorise L .

$$\frac{d^2 y}{dx^2} + 5 \frac{dy}{dx} + 6y = 0$$

2. Verify that L annihilates the general solution of .

Answer: